

ROSWELL, NM, USA

WMO: 722680

Lat: 33.308N

Lon: 104.508W

Elev: 3649

StdP: 12.86

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23009

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	17.7	21.6	-2.2	5.6	43.8	1.5	6.8	45.3	31.1	58.7	26.7	55.8	4.6	0	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.8	101.9	64.2	99.2	64.7	96.8	64.7	70.8	86.8	69.6	86.0	68.9	85.3	12.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	114.3	74.2	65.9	109.5	73.7	64.7	104.9	73.6	36.9	86.5	36.0	85.9	35.2	85.2	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.0	23.4	19.4	DB	9.7	106.5	6.2	2.3	5.3	108.1	1.7	109.5	-1.8	110.7	-6.3	112.4
			WB	7.9	73.9	5.7	3.9	3.8	76.7	0.4	79.0	-2.8	81.2	-7.0	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	41.7	46.6	54.3	62.1	71.3	80.0	81.8	80.3	73.5	62.3	50.5	41.4	(6)
(7)		DBStd	16.17	8.22	8.85	8.29	7.62	6.96	4.68	3.97	4.14	5.69	7.52	7.94	8.52	(7)
(8)		HDD50	871	274	150	54	10	0	0	0	0	0	9	89	285	(8)
(9)		HDD65	3051	722	516	337	137	26	0	0	0	9	137	437	730	(9)
(10)		CDD50	5336	17	55	188	373	659	899	986	940	705	391	103	20	(10)
(11)		CDD65	2042	0	0	7	51	220	449	521	475	264	54	1	0	(11)
(12)		CDH74	26481	9	70	381	1170	3176	6001	6435	5461	2815	867	91	5	(12)
(13)		CDH80	14091	0	10	88	430	1641	3568	3687	2991	1352	313	11	0	(13)
(14)	Wind	WSAvg	8.5	7.4	8.7	9.6	10.7	10.0	10.0	8.4	7.6	7.8	7.7	7.3	7.1	(14)
(15)	Precipitation	PrecAvg	12.4	0.4	0.4	0.5	0.5	1.0	1.5	1.9	2.1	1.7	1.1	0.6	0.6	(15)
(16)		PrecMax	24.8	1.5	2.0	2.8	2.9	4.6	5.0	6.9	6.5	6.6	5.9	3.0	3.1	(16)
(17)		PrecMin	2.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.7	0.4	0.5	0.6	0.7	1.1	1.4	1.8	1.6	1.6	1.5	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.5	81.1	86.9	92.8	100.4	105.7	104.4	103.0	98.1	91.2	81.5	74.1	(19)
(20)			MCWB	49.8	52.2	53.9	56.2	60.2	62.9	66.0	66.2	66.0	60.4	53.8	49.9	(20)
(21)		2%	DB	69.5	75.7	82.3	88.4	96.9	102.7	100.6	99.0	94.6	87.3	76.4	68.3	(21)
(22)			MCWB	47.6	50.1	52.5	54.8	59.4	62.8	66.2	66.7	64.3	59.5	53.0	47.9	(22)
(23)		5%	DB	64.3	70.9	78.5	85.1	93.6	99.8	98.1	96.4	91.7	83.5	72.0	63.2	(23)
(24)			MCWB	45.1	47.9	51.3	54.4	58.7	63.0	66.4	66.8	64.0	58.5	51.3	45.5	(24)
(25)		10%	DB	58.9	65.4	73.7	81.5	89.6	96.3	95.6	93.7	88.3	78.9	67.1	57.7	(25)
(26)			MCWB	42.9	45.6	49.6	53.1	58.0	63.2	66.4	66.8	64.0	57.7	49.2	43.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	54.0	58.8	61.4	67.1	71.9	72.5	71.5	71.0	68.4	59.3	52.1	(27)
(28)			MCD B	70.3	75.6	74.6	76.8	83.9	88.6	88.2	87.1	85.4	78.1	69.0	66.4	(28)
(29)		2%	WB	48.8	51.4	55.3	59.2	65.1	69.4	70.7	70.4	69.1	65.6	55.9	49.4	(29)
(30)			MCD B	65.7	71.9	74.4	75.6	82.8	87.2	87.3	86.3	82.7	74.4	69.0	65.7	(30)
(31)		5%	WB	46.2	49.1	52.9	57.2	63.3	68.1	69.6	69.4	67.9	62.7	53.1	46.5	(31)
(32)			MCD B	62.0	68.7	72.9	76.6	81.8	86.1	86.5	85.7	80.8	74.5	68.4	61.6	(32)
(33)		10%	WB	43.7	46.5	50.8	55.1	61.6	66.9	68.8	68.7	66.6	60.1	50.7	43.9	(33)
(34)			MCD B	58.3	63.6	70.5	76.1	80.3	85.0	85.8	85.1	79.7	73.3	65.0	56.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	29.0	30.1	30.9	31.4	30.3	29.7	25.8	25.2	25.7	27.7	28.6	27.3	(35)
(36)			MCDBR	38.7	39.9	39.1	37.5	36.5	35.5	29.9	29.1	31.3	35.0	36.6	36.5	(36)
(37)			MCWBR	21.2	20.2	18.2	16.2	14.8	12.2	8.9	8.3	10.7	14.6	18.4	20.0	(37)
(38)		5% WB	MCD BR	35.9	37.3	34.9	32.4	30.4	28.2	24.7	24.5	24.0	25.8	31.5	33.2	(38)
(39)			MCWBR	20.4	19.4	17.1	15.5	13.4	9.9	7.5	7.0	9.0	12.3	16.7	18.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.250	0.259	0.278	0.308	0.327	0.354	0.384	0.364	0.335	0.292	0.268	0.256	(40)
(41)		taud		2.615	2.596	2.535	2.432	2.436	2.400	2.364	2.412	2.451	2.576	2.576	2.591	(41)
(42)		Ebn at Noon		314	321	319	311	303	294	284	289	295	304	304	305	(42)
(43)		Edh at Noon		24	27	31	36	36	38	39	36	34	27	24	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		1099	1402	1799	2216	2403	2457	2307	2102	1816	1465	1154	960	(44)
(45)		RadStd		92	120	118	134	129	112	110	84	114	136	95	77	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.39	N/A	N/A	N/A	N/A	+245	N/A	(46)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page